

College of Industrial Technology
King Mongkut's University of Technology North Bangkok

Seat No.

Midterm Examination of Semester 1

Year: 2019

Subject: 394173 Mathematics 3

Section: 15-18

Date: 14 August 2019

Time: 10.00-12.00

Name: _____ ID: _____ Class: _____

Directions: The test is designed to measure your comprehension. The test is divided into one section. There will be 9 pages (including this page) and they are worth 80 points.

- Answer the questions on this test papers.
- Books, documents and lecture notes are not allowed.
- You must be in the room for one hour after the exam is started and, while taking the exam, you cannot go out except in an emergency case.
- Before leaving, make sure you do not bring this test outside.
- Do not use any electronic communication device.
- Calculators can be used in this test.

Now begin the test.

Cheating in the exam is considered an extremely serious offence which will result in expulsion from the University.

2. 2.1 Let $|(1 - i)^2(-3 - \sqrt{7}i)z^6| = 64$. Find $z \cdot \overline{z}$. (5 marks)

2.2 Let x be the complex number. Find the roots of equation

$$x^3 - (7 + 5i)x^2 + (12 + 35i)x - 60i = 0. \quad (5 \text{ marks})$$

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4. Let z_0, z_1 and z_2 be 3th roots of $-1 + \sqrt{3}i$, find $\left| \frac{z_1 \cdot z_2}{z_0^2} \right|$. (10 marks)

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6. 6.1 If the matrix $A = \begin{bmatrix} 0 & 0 & a \\ b & 0 & 0 \\ 0 & c & 0 \end{bmatrix}$ where a, b, c are positive numbers such that

$abc = 1$, evaluate the value of $\det(A^2 + A + I)$. (5 marks)

6.2 Given the matrices $A = \begin{bmatrix} 2 & 0 \\ -1 & 1 \end{bmatrix}$ and $B = \begin{bmatrix} a & b \\ c & d \end{bmatrix}$

where $B = A^{-1}BA$, find the value of $a + b + c + d$. (5 marks)

8. Use Cramer's rule to solve the system of three equations with three unknown

$$2x + 3y - \frac{1}{z} = 1$$

$$4x + y - \frac{3}{z} = 11 \quad (10 \text{ marks})$$

$$3x - 2y + \frac{5}{z} = 21$$

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